

Fraudulent Insurance Claim Detection

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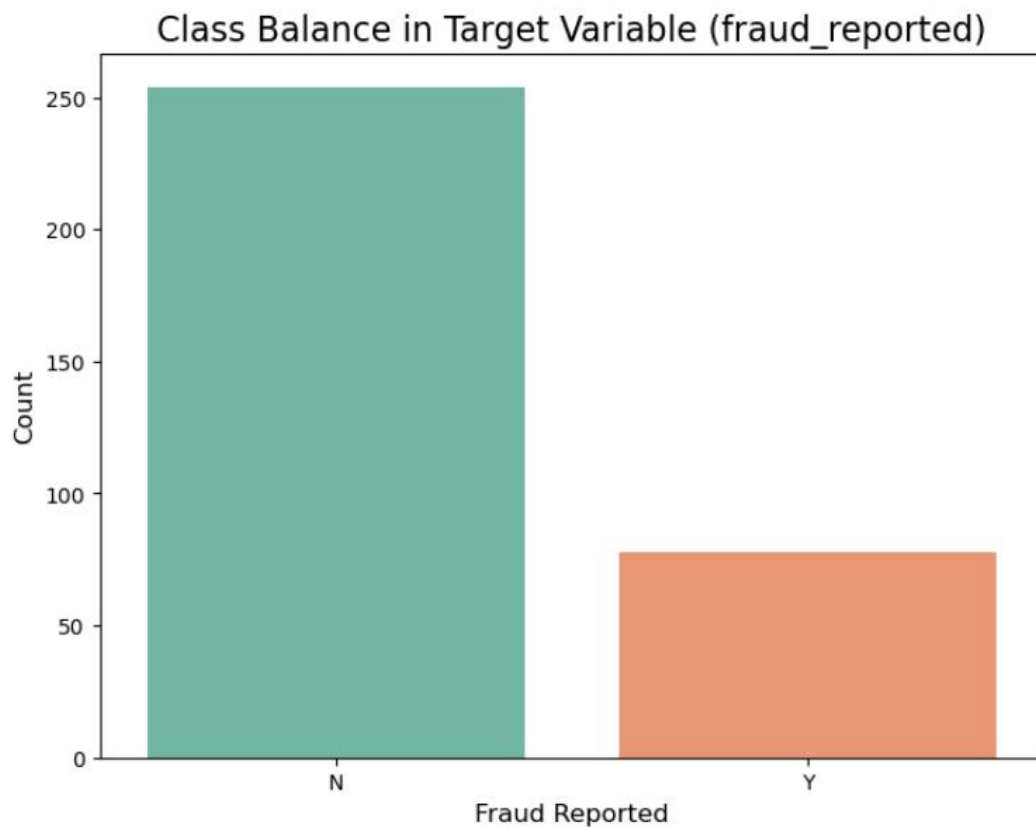
Introduction & Objective

The report examines an insurance claims dataset to uncover fraudulent activities through exploratory analysis and predictive modeling. The primary goal is to recognize fraud patterns, analyze claim behaviors, and identify the key factors associated with suspicious claims.

Dataset Overview

The dataset comprises 1,000 insurance claims. Each is described by 40 attributes including customer demographics, vehicle details, policy information, incident-related data, and a target field labeled 'fraud_reported' that specifies whether the claim was marked as fraudulent ('Y') or not ('N').

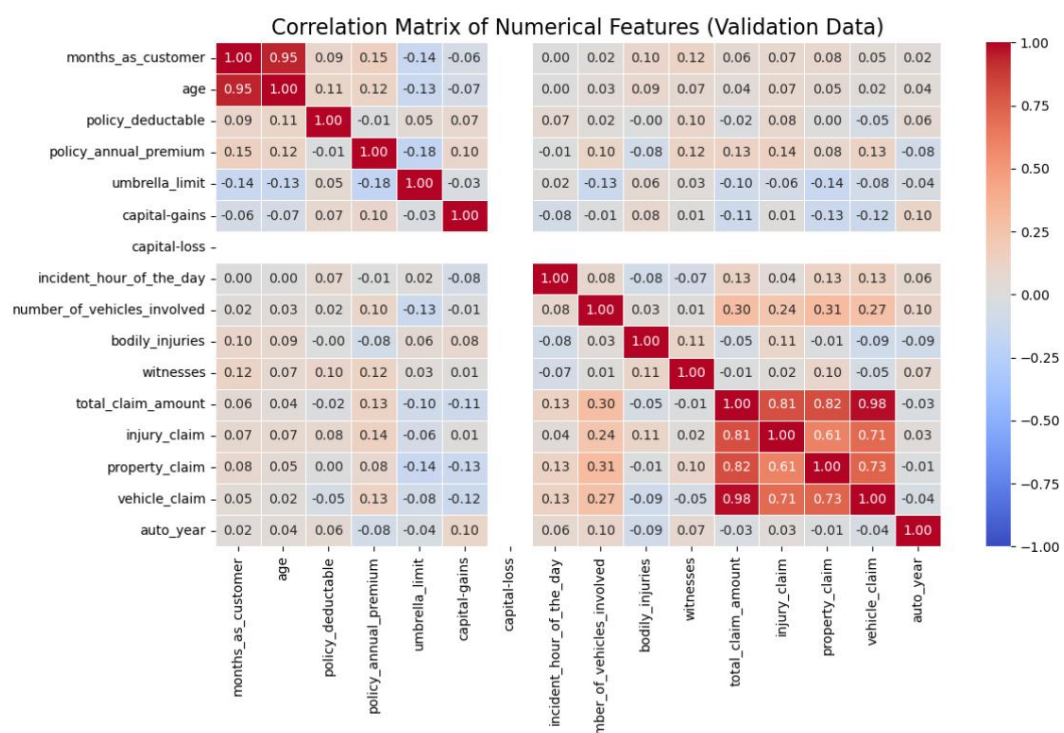
Exploratory Data Analysis (EDA)



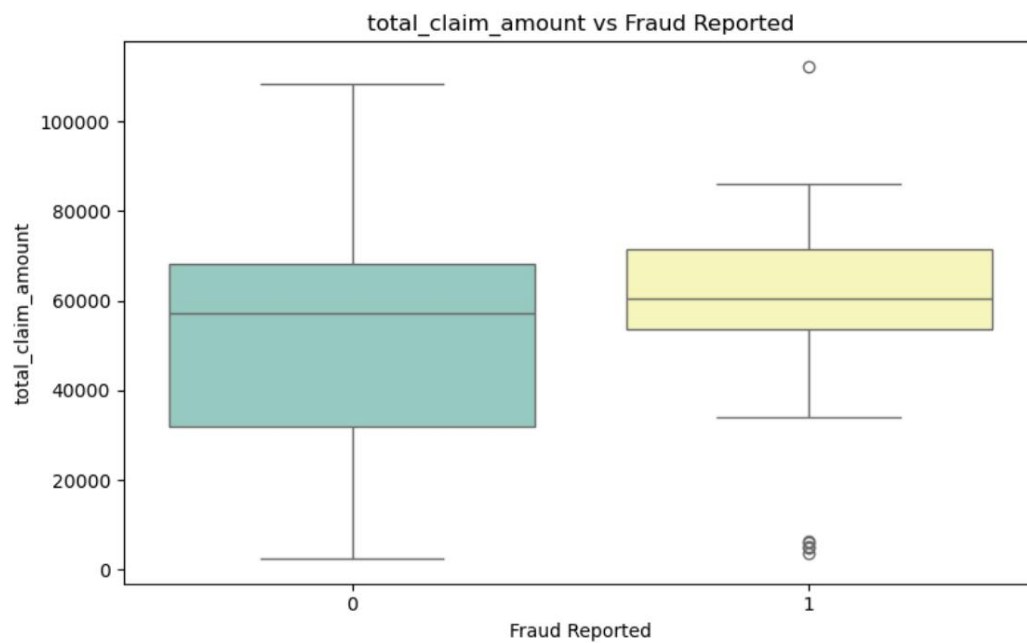
Out of a total of 1,000 records:

- 231 claims were identified as fraudulent (23.1%)
- 769 claims were identified as non-fraudulent (76.9%)

This suggests that nearly 1 in 4 claims could potentially be fraudulent, which is a significant finding.



BoxPlot:



The boxplot shows the differences in total claim amounts between fraud-reported and non fraud cases, with fraud-reported claims showing a higher median. Non-fraud cases have a wider range of claim amounts, while both categories feature outliers. This visualization offers valuable insights for identifying distinct trends in fraud detection.

Detailed Statistical Insights

Class Imbalance Identified:

The insurance claims dataset displays a significant imbalance, with non-fraudulent claims occurring more frequently than fraudulent ones, potentially skewing model predictions toward the majority class.

Feature Correlations:

Features such as age, policy number, and total_claim_amount are provided. It has been noticed that the total_claim_amount tends to correlate more strongly with fraudulent claims.

Distribution Plots:

Different patterns are observed in the distributions of claim amounts and customer lifetime value for fraud and non-fraud related cases.

Data Processing & Modelling Summary

Data Cleaning:

Handled Missing values appropriately.

Categorical features were encoded using one-hot encoding.

Feature Selection:

Irrelevant or redundant features were dropped.

Feature importance was assessed using tree-based models.

Modelling Approaches:

Multiple models tested: Logistic Regression, Decision Tree, Random Forest, XGBoost. XGBoost achieving the best performance after hyperparameter optimization via grid search.

Evaluation Metrics:

Metrics such as F1 Score, Accuracy, Precision, and Recall were utilized.

Confusion matrices showed XGBoost balanced precision and recall effectively for fraud detection.

Conclusion & Recommendations

To conclude this, XGBoost is considered to be the best performing model. This model has achieved high precision and recall which is particularly critical for fraud detection where false negatives are costly.

In this model, techniques like SMOTE and other oversampling methods were used to handle class imbalance. These approaches help improve recall for the minority class (fraud), ensuring more effective mitigation of the imbalance.

Improving feature engineering might include analysing domain-specific variables, like patterns in claim timing or the history of claims per customer. These elements could provide additional predictive value, strengthening the model's accuracy and effectiveness.